
From Letters to Networks: Historical Network Analysis of Dutch Brazil through the ePistolarium

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Abstract

The historiography of Dutch Brazil (1630–1654) has long centered on territorial, political, and military dimensions. Particularly the figure of Johann Moritz von Nassau-Siegen as a symbol of urban progress and enlightened colonial governance, a framing that has also shaped Brazilian school curricula. This paper demonstrates that knowledge about Dutch Brazil circulated extensively across multiple spheres of intellectual and political life in the Dutch Republic, and that network analysis of early modern correspondence reveals who the central actors in that circulation were. Our primary source is the ePistolarium platform (Ravenek, van den Heuvel, and Gerritsen 2017), hosting approximately 20,000 letters exchanged among 17th-century scholars in the Dutch Republic. Search terms were derived from key works on natural knowledge produced during the Dutch Brazil period, including *Historia Naturalis Brasiliae* (1648) and *Rerum per octennium in Brasilia* (1647), yielding a corpus of 281 letters after deduplication. Strikingly, only 12 were sent or received in Brazil, 7 of them by Nassau himself, confirming that colonial debate was largely conducted within European scholarly and political circles.

The research was conducted by a student team in a scientific initiation program at the secondary school level at Instituto Federal do Sudeste de Minas Gerais. The team developed two Python-based tools: a converter transforming ePistolarium JSON exports into GEXF format for Gephi, and a term-verification interface eliminating query redundancies. A GIS visualization layer using Leaflet was also integrated for geographic mapping. Network analysis identified Hugo de Groot (Grotius), Constantijn Huygens, and Caspar Barlaeus as the central nodes, with Barlaeus bridging the two main clusters in the correspondence network.

Close reading of the letters reveals that Brazil was not a peripheral concern but a transversal theme across multiple spheres of Dutch elite discourse. Grotius engaged with Brazil as a strategic and diplomatic problem, discussing it as an obstacle to peace with Spain and assessing the West India Company's successes and failures from Paris and The Hague. Huygens served as Nassau's primary interlocutor in the Republic, acting as a channel between the governor and the Prince of Orange and relaying detailed accounts of colonial affairs. Barlaeus, commissioned to write the official history of Nassau's government, transformed colonial experience into metropolitan academic glorification. Beyond these central figures, Brazil appeared incidentally across letters on navigation, oceanography, and European military affairs, indicating that the colony functioned as a reference point across scientific, diplomatic, commercial, and intellectual conversations. Colonial governance was therefore not delegated to the Company or the governor alone, but filtered through a transatlantic

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circuit of power and knowledge in which Dutch intellectual circles actively mediated, supervised, and legitimized the colonial project.

These findings connect to growing scholarship on natural knowledge in Dutch Brazil, including recent work on Georg Marggrafe's astronomical observations (Matsuura and Zuidervaart 2022) and the ERC-funded BRASILIAE research project on indigenous knowledge and *Historia Naturalis Brasiliae* (Ariese 2018–2023). As Graham, Milligan, and Weingart (2016) argue, network perspectives make "the invisible visible", here revealing actors and dynamics absent from standard historical and educational accounts. This paper shows that HNR methods are productive even at the secondary school level, offering young researchers access to sophisticated historiographical questions through computational tools.

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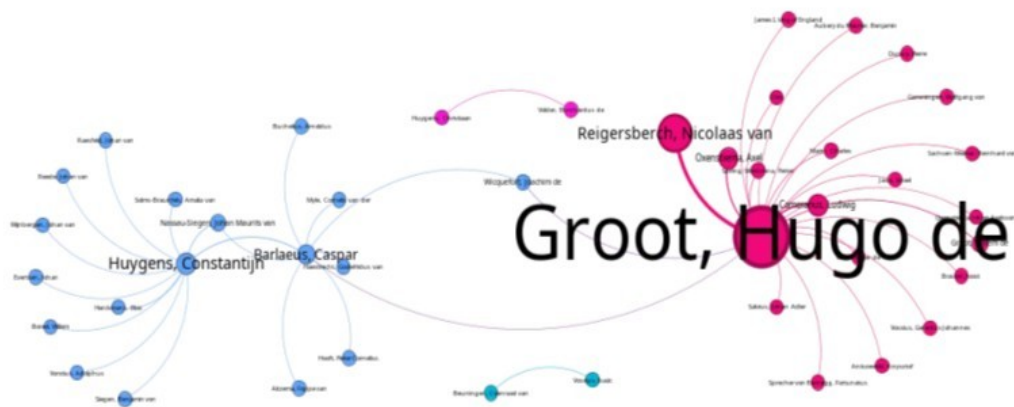


Figure 1. Correspondence network from the ePistolarium corpus (281 letters, 1620–1650). Node size proportional to weighted degree. Visualization produced in Gephi.